

DATA ANALYTICS

Task Presentation

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ACKNOWLEDGEMENT

I would like to express my gratitude to ShadowFox for giving me the opportunity to present the tasks I have completed during this internship.

I would also like to extend my thanks to my coordinator, Mr. Aakash, for guiding me throughout this experience. I acknowledge that the tasks were completed independently by me. Additionally, I utilized the Superstore Sales dataset and Google Sheets pivot tables for performing data cleaning and generating strategic business insights.

BEGINNER LEVEL

Task: Superstore Sales Data Analysis & Visualization Techniques using Pivot Tables

- Analyzed Superstore Sales dataset (9,994 orders, \$2.3M Total Revenue) using Pivot Tables, Matplotlib, and Seaborn ,GoogleSheet .
- Created key pivot table visualizations:
 - Line plot (Monthly Revenue Trend)
 - Scatter plot (Sales Amount vs Profit Correlation)
 - Bar chart (Category-wise Sales Revenue)
 - Histogram (Order Value Distribution)
 - Pie chart (Regional Sales Contribution)
 - Box plot (Profit Spread by Customer Segment)
 - Pair plot / Heatmap (Category vs Region Profitability)

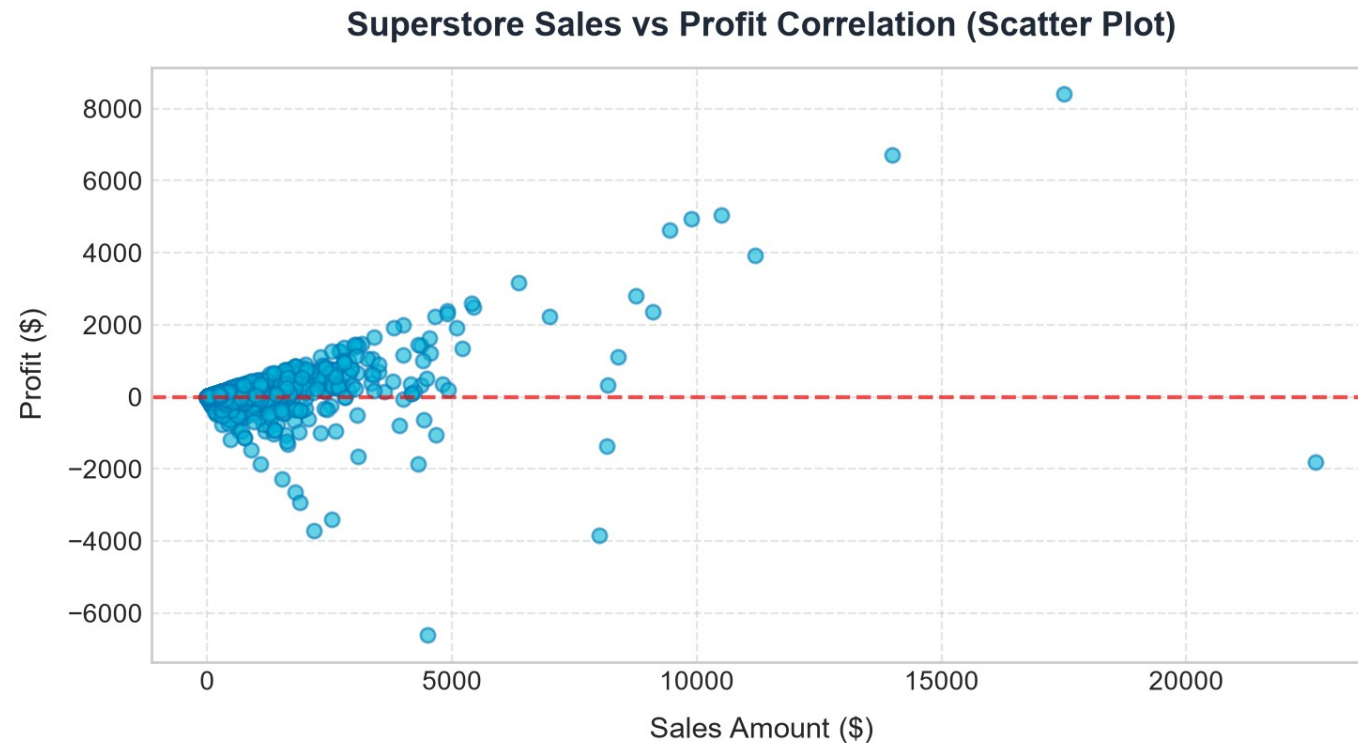
Line Plot

- Description: Line plots represent continuous Superstore sales trends across time, tracking monthly growth trajectory.
- Use Case: Analyzing monthly Superstore revenue movement. Highlights steady sales growth with Q4 seasonal demand peaks.



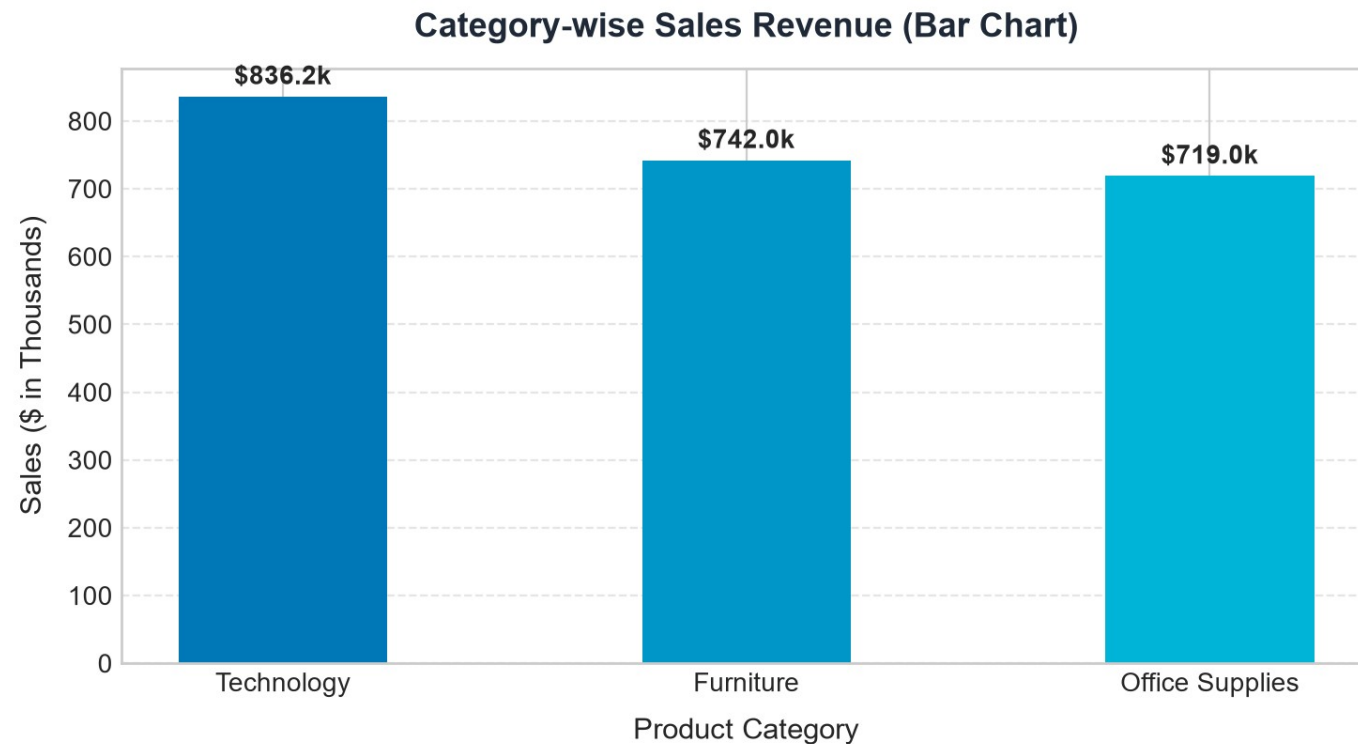
Scatter Plot

- Description: Scatter plots display order-level transactions to analyze correlation between Superstore Sales and Profit.
- Use Case: Evaluating profit scaling per order. Confirms overall positive margin correlation while revealing loss-making heavy discount orders.



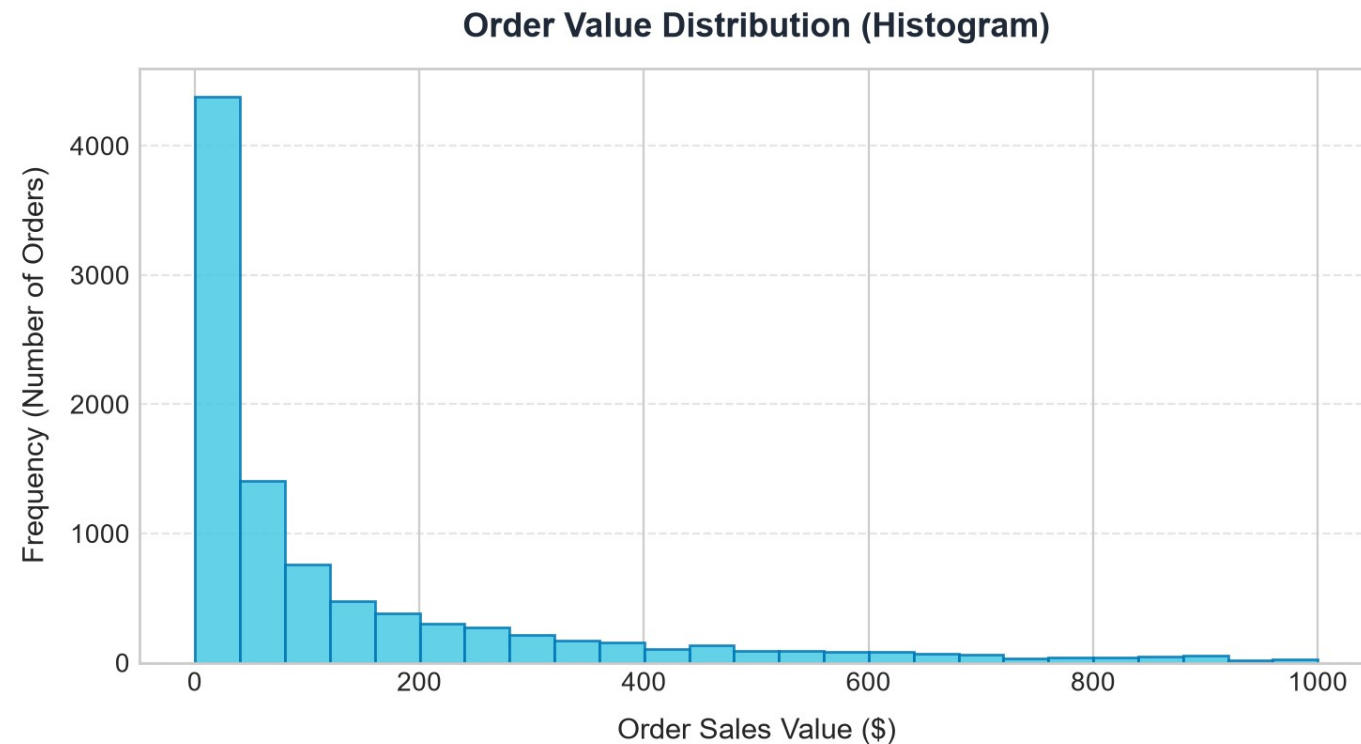
Bar Plot

- Description: Bar charts compare aggregated revenue across product categories: Technology, Furniture, and Office Supplies.
- Use Case: Comparing category performance. Technology led total revenue (\$836k), followed by Furniture (\$742k) and Office Supplies (\$719k).



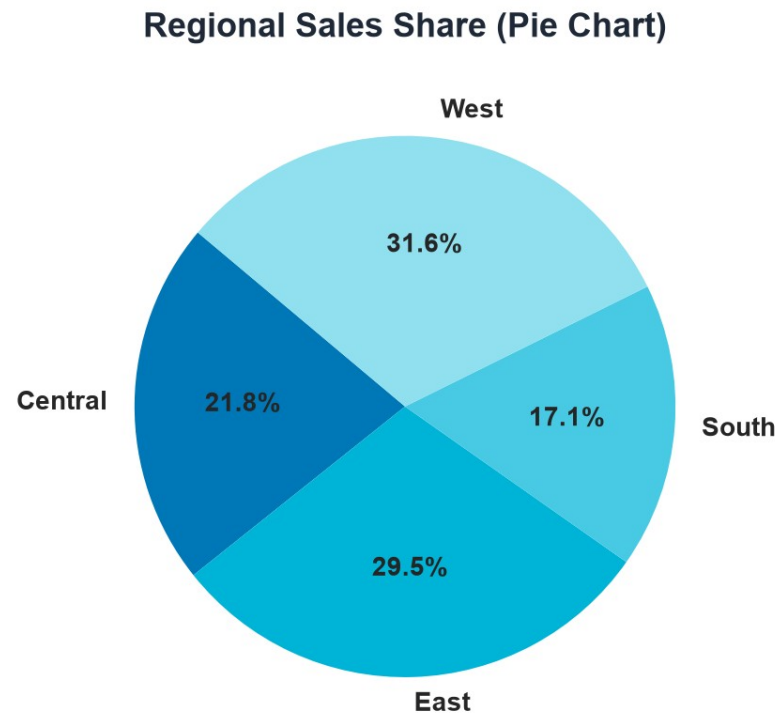
Histogram

- Description: Histograms illustrate order size frequency distributions across equal sales value bins.
- Use Case: Analyzing Superstore order sizes. Over 80% of orders fall under \$500, informing bundled promotional strategy.



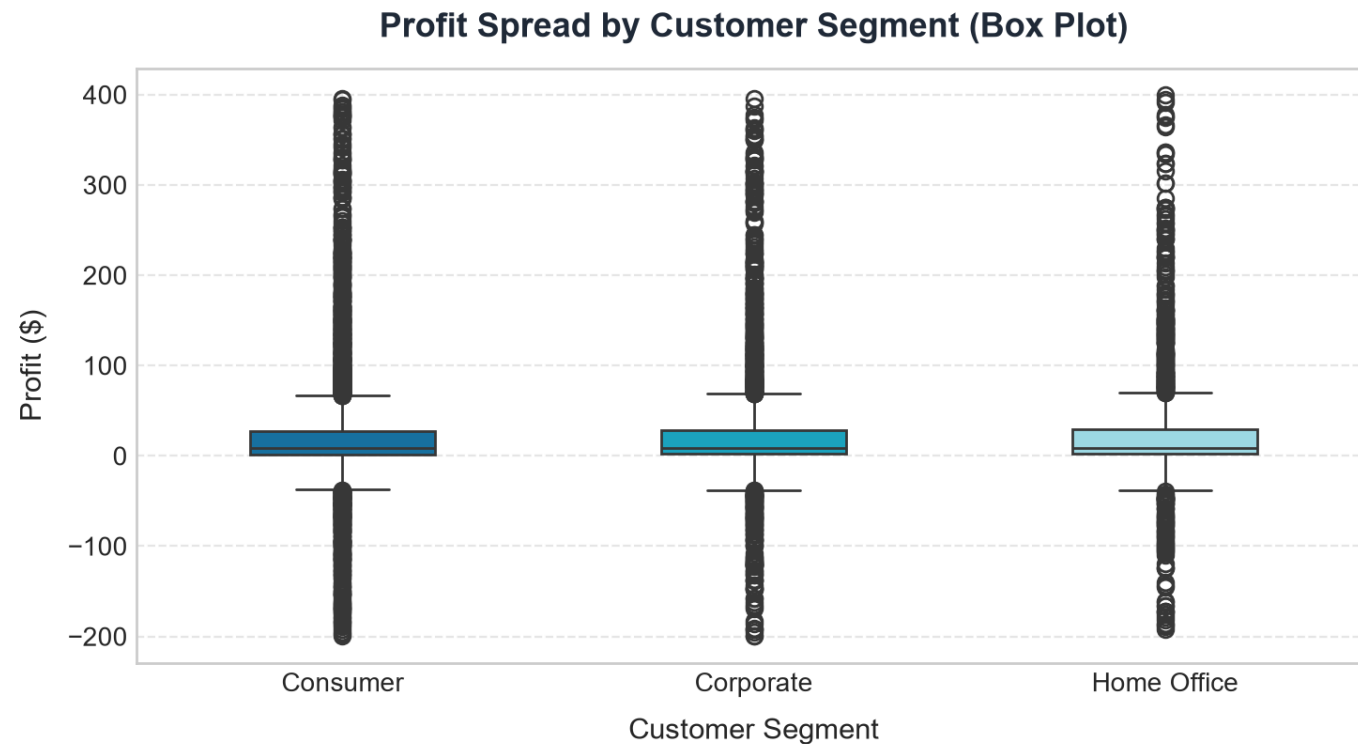
Pie Chart

- Description: Pie charts illustrate percentage share of Superstore revenue contributed by each geographical region.
- Use Case: Evaluating regional market share: West region leads with 31.6%, East with 29.5%, Central with 21.8%, and South with 17.1%.



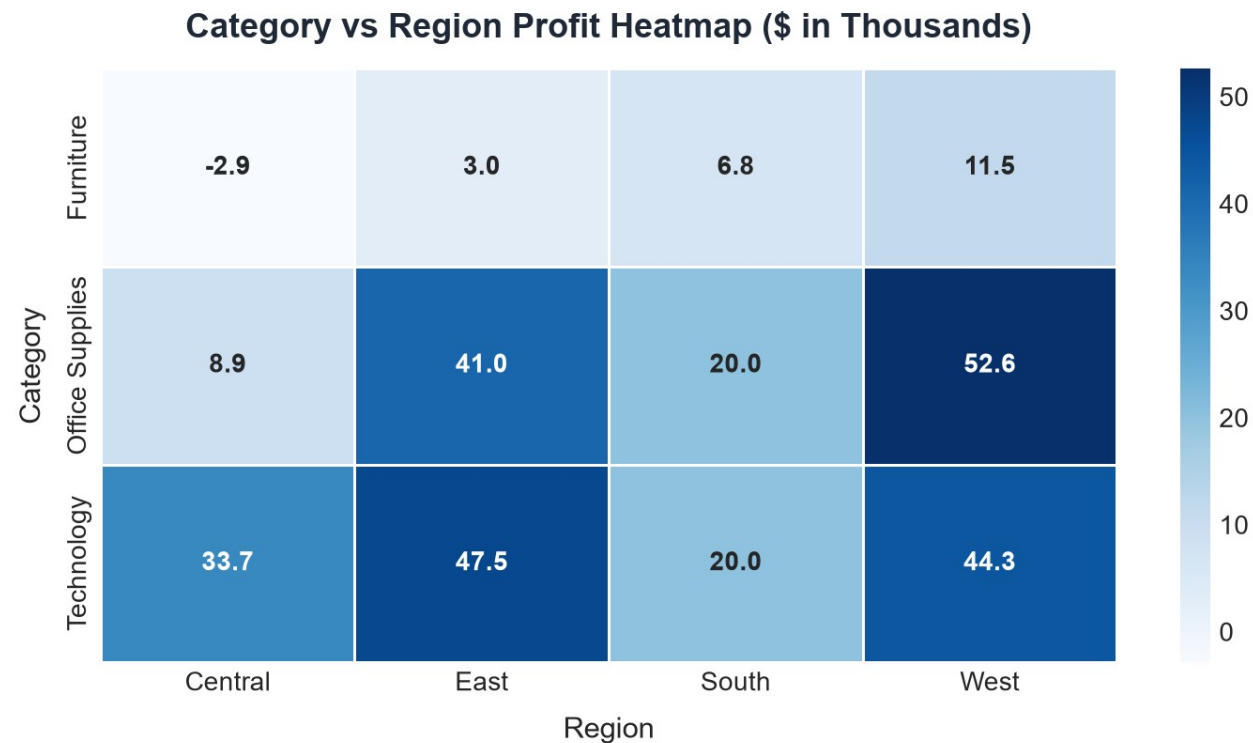
Box Plot

- Description: Box plots compare profit distributions (min, Q1, median, Q3, max) across Consumer, Corporate, and Home Office customer segments.
- Use Case: Assessing segment profitability spread. Corporate segment yields higher median order profit compared to Consumer retail orders.



Pair Plot / Pivot Matrix

- Description: Cross-tabulated pivot matrix displaying total profit (\$ in Thousands) aggregated by Product Category and Region.
- Use Case: Pinpointing regional profit drivers. Technology in the East (\$47.4k profit) and West (\$44.3k profit) generated highest margins.



CONCLUSION



This internship provided valuable hands-on experience analyzing the Superstore Sales dataset using Google Sheets pivot tables and Python visualization tools. By completing the tasks outlined, I developed strong skills in data cleaning, KPI aggregation, pivot modeling, and presenting data-driven business insights.

THANK YOU